

Package #165

apply a magnetorheological (MR) fluid coating to the rail surface and energize it with a magnetic field during service to generate a compressive stress field at potential crack tips, thereby inducing a crack■tip shielding effect under cyclic loading

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'fatigue fracture' concerning railway rail. User-submitted engineering problem (session input): "Why do rails fracture in service under fatigue loading?". [cod_optimade] retrieval status EMPTY — provider outcome recorded, not treated as absence (Art. XXI.3). [arxiv] 5 records retrieved live (2010-2020). Illustrative records: 'How we can control the crack to propagate along the specified path feasibly?'; 'Steady-state propagation of a Mode III crack in couple stress elastic materials'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: crack tip shielding effect that correlates crack growth rate with the loading force range ΔF in addition to the stress■intensity range ΔK

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.